

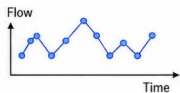
1

Inputs

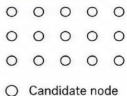
Traffic network



Historical traffic states



Sensor budget & candidate nodes



2

Recoverability-Driven
Bilevel OptimizationUpper
levelChoose sensor layout z
(subset of candidate nodes,
 $|z| = \text{budget } k$)Lower
levelReconstruct hidden traffic
states x using a
transparent GLS/MAP
estimator

Objective components

Reconstruction
errorPosterior
uncertainty

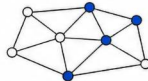
Tail risk

Spatial
redundancy penaltyDeterministic solver /
exchange refinement

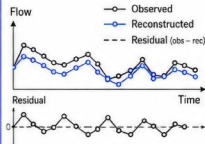
3

Selected layout &
evaluation

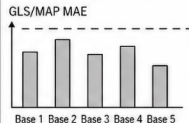
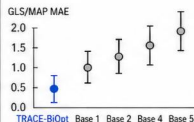
Selected sensor layout



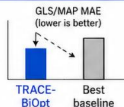
● Selected sensor

Held-out reconstruction
evaluationMetric:
GLS/MAP MAE

4

Comparison &
claim boundaryPre-registered
non-BiOpt baselinesCurrent-best
comparison

i Baselines are used only
for held-out comparison,
not inside the solver.



Output: a recoverability-aware sensor deployment with
lower held-out hidden-state reconstruction error.